

Q.9

$$\begin{aligned}
 a) \quad u_{i+1,j+1} &= u_{i,j} + (\Delta x) u_x + (\Delta y) u_y + \frac{1}{2!} \left[(\Delta x)^2 u_{xx} + (\Delta y)^2 u_{yy} + 2(\Delta x)(\Delta y) u_{xy} \right] \\
 &+ \frac{1}{3!} \left[(\Delta x)^3 u_{xxx} + (\Delta y)^3 u_{yyy} + 3(\Delta x)^2(\Delta y) u_{xxy} + 3(\Delta x)(\Delta y)^2 u_{xyy} \right] \\
 &+ \frac{1}{4!} \left[(\Delta x)^4 u_{xxxx} + (\Delta y)^4 u_{yyyy} + 4(\Delta x)^3(\Delta y) u_{xxx} + 6(\Delta x)^2(\Delta y)^2 u_{xxyy} + 4(\Delta x)(\Delta y)^3 u_{xyyy} \right] \\
 &+ O[(\Delta x)^5, (\Delta y)^5]
 \end{aligned}$$

$$\begin{aligned}
 u_{i-1,j-1} &= u_{i,j} - (\Delta x) u_x - (\Delta y) u_y + \frac{1}{2!} \left[(\Delta x)^2 u_{xx} + (\Delta y)^2 u_{yy} + 2(\Delta x)(\Delta y) u_{xy} \right] \\
 &+ \frac{1}{3!} \left[-(\Delta x)^3 u_{xxx} - (\Delta y)^3 u_{yyy} - 3(\Delta x)^2(\Delta y) u_{xxy} - 3(\Delta x)(\Delta y)^2 u_{xyy} \right] \\
 &+ \frac{1}{4!} \left[(\Delta x)^4 u_{xxxx} + (\Delta y)^4 u_{yyyy} + 4(\Delta x)^3(\Delta y) u_{xxx} + 6(\Delta x)^2(\Delta y)^2 u_{xxyy} + 4(\Delta x)(\Delta y)^3 u_{xyyy} \right] \\
 &+ O[(\Delta x)^5, (\Delta y)^5]
 \end{aligned}$$

$$\begin{aligned}
 u_{i-1,j+1} &= u_{i,j} - (\Delta x) u_x + (\Delta y) u_y + \frac{1}{2!} \left[(\Delta x)^2 u_{xx} + (\Delta y)^2 u_{yy} - 2(\Delta x)(\Delta y) u_{xy} \right] \\
 &+ \frac{1}{3!} \left[-(\Delta x)^3 u_{xxx} + (\Delta y)^3 u_{yyy} + 3(\Delta x)^2(\Delta y) u_{xxy} - 3(\Delta x)(\Delta y)^2 u_{xyy} \right] \\
 &+ \frac{1}{4!} \left[(\Delta x)^4 u_{xxxx} + (\Delta y)^4 u_{yyyy} - 4(\Delta x)^3(\Delta y) u_{xxx} + 6(\Delta x)^2(\Delta y)^2 u_{xxyy} - 4(\Delta x)(\Delta y)^3 u_{xyyy} \right] \\
 &+ O[(\Delta x)^5, (\Delta y)^5]
 \end{aligned}$$

$$\begin{aligned}
 u_{i+1,j-1} &= u_{i,j} + (\Delta x) u_x - (\Delta y) u_y + \frac{1}{2!} \left[(\Delta x)^2 u_{xx} + (\Delta y)^2 u_{yy} - 2(\Delta x)(\Delta y) u_{xy} \right] \\
 &+ \frac{1}{3!} \left[(\Delta x)^3 u_{xxx} - (\Delta y)^3 u_{yyy} - 3(\Delta x)^2(\Delta y) u_{xxy} + 3(\Delta x)(\Delta y)^2 u_{xyy} \right] \\
 &+ \frac{1}{4!} \left[(\Delta x)^4 u_{xxxx} + (\Delta y)^4 u_{yyyy} - 4(\Delta x)^3(\Delta y) u_{xxx} + 6(\Delta x)^2(\Delta y)^2 u_{xxyy} - 4(\Delta x)(\Delta y)^3 u_{xyyy} \right] \\
 &+ O[(\Delta x)^5, (\Delta y)^5]
 \end{aligned}$$

$$\begin{aligned}
 u_{i+1,j+1} - u_{i+1,j-1} - u_{i-1,j+1} + u_{i-1,j-1} &= \frac{1}{2!} \left[8(\Delta x)(\Delta y) u_{xy} \right] + \frac{1}{4!} \left[16(\Delta x)^3(\Delta y) u_{xxx} + 16(\Delta x)(\Delta y)^3 u_{xyy} \right] \\
 &+ O[(\Delta x)^5, (\Delta y)^5]
 \end{aligned}$$

$$\therefore \frac{\partial^2 u(i,j)}{\partial x \partial y} \approx \frac{u_{i+1,j+1} - u_{i+1,j-1} - u_{i-1,j+1} + u_{i-1,j-1}}{4(\Delta x)(\Delta y)} - \left[\frac{1}{6} (\Delta x)^2 u_{xxx} + \frac{1}{6} (\Delta y)^2 u_{xyy} \right]$$

$$\therefore \text{the Leading-order Error is } \frac{1}{6} \left[(\Delta x)^2 \frac{\partial^3 u(i,j)}{\partial x^3 \partial y} + (\Delta y)^2 \frac{\partial^3 u(i,j)}{\partial x \partial y^3} \right]$$

$$b) u(x, y) = y^4 \sin x$$

$$\frac{\partial u}{\partial x} = y^4 \cos x$$

$$\frac{\partial u}{\partial y} = 4y^3 \sin x$$

$$\frac{\partial^2 u}{\partial x \partial y} = 4y^3 \cos x$$

Exact Value :

$$\therefore \frac{\partial^2}{\partial x \partial y} u(0, 1) = 4$$

$$u_{i+1, j+1} = u(0.1, 1.1) = 1.1^4 \sin(0.1) = 0.146166$$

$$u_{i+1, j-1} = u(0.1, 0.9) = 0.9^4 \sin(0.1) = 0.065501$$

$$u_{i-1, j+1} = u(-0.1, 1.1) = 1.1^4 \sin(-0.1) = -0.146166$$

$$u_{i-1, j-1} = u(-0.1, 0.9) = 0.9^4 \sin(-0.1) = -0.065501$$

Finite-difference Value :

$$\frac{\partial^2}{\partial x \partial y} u(0, 1) = \frac{u_{i+1, j+1} - u_{i+1, j-1} - u_{i-1, j+1} + u_{i-1, j-1}}{4(0.1)(0.1)} = 4.03325$$

c) The compute error in b) ≈ 0.03325

$$\text{The Leading-order error} = \frac{1}{6} \left[(\Delta x)^2 \frac{\partial^4 u}{\partial x^3 \partial y} + (\Delta y)^2 \frac{\partial^4 u}{\partial x \partial y^3} \right]$$

$$\therefore \frac{\partial^4 u}{\partial x^3 \partial y}(x, y) = -4y^3 \cos x \xrightarrow[y=1]{x=0} -4$$

$$\frac{\partial^4 u}{\partial x \partial y^3}(x, y) = 24y \cos x \xrightarrow[y=1]{x=0} 24$$

$$\therefore \frac{1}{6} \left[(0.1)^2 \frac{\partial^4 u}{\partial x^3 \partial y}(0, 1) + (0.1)^2 \frac{\partial^4 u}{\partial x \partial y^3}(0, 1) \right] = \frac{1}{6} \times 0.01 \times 20 = \frac{1}{30} \approx 0.0333$$

$\therefore$ Indeedly the error depend on the Leading-order error term.

Q10.

$$a) \frac{du}{dt} = \frac{10t}{t^5 + c} - \frac{25t^6}{(t^5 + c)^2} = \frac{2u}{t} - t^2 u^2$$

$$b) \frac{DF}{Dt} = \frac{\partial F}{\partial u} \frac{du}{dt} + \frac{\partial F}{\partial t}$$

$$F = \frac{du}{dt} = \frac{2u}{t} - t^2 u^2 \quad \therefore \frac{\partial F}{\partial u} = \frac{2}{t} - 2t^2 u \quad \frac{\partial F}{\partial t} = -\frac{2u}{t^2} - 2tu^2$$

$$\therefore \frac{DF}{Dt} = \left(\frac{2}{t} - 2t^2 u \right) \left(\frac{2u}{t} - t^2 u^2 \right) - \frac{2u}{t^2} - 2tu^2$$

$$= \frac{4u}{t^2} - 2tu^2 - 4tu^2 + 2t^4 u^3 - \frac{2u}{t^2} - 2tu^2$$

$$= \frac{2u}{t^2} - 8tu^2 + 2t^4 u^3$$

$$c) \frac{d^2 u}{dt^2} = \frac{d}{dt} \left(\frac{2u}{t} - t^2 u^2 \right) = -\frac{2u}{t^2} + \frac{2}{t} \frac{du}{dt} - 2tu^2 - t^2 \frac{du^2}{dt}$$

$$= -\frac{2u}{t^2} + \frac{2}{t} \left(\frac{du}{dt} \right) - 2tu^2 - 2ut^2 \left(\frac{du}{dt} \right)$$

$$= \frac{2u}{t^2} - 8tu^2 + 2t^4 u^3$$

$$\therefore \frac{DF}{Dt} = \frac{du}{dt}$$